

Project README & Documentation

Task 5: Exploratory Data Analysis (EDA) on E-Commerce Sales Dataset

Python 3.x

Pandas

Matplotlib

Seaborn

VS Code Jupyter

15,030

RECORDS

25

FEATURES

6

SALES CHANNELS

\$5.28M

TOP CATEGORY SALES

1. Project Overview

This project delivers a comprehensive Exploratory Data Analysis (EDA) performed in Visual Studio Code using Jupyter Notebooks. The study investigates consumer purchasing patterns, sales channel diversification, pricing elasticity, and data anomalies across global market regions.

2. Dataset Specifications

Attribute	Details
Source File	salesdataset.csv
Dimension	15,030 rows × 25 columns
Key Categorical Fields	Region, Country, SalesChannel, ProductCategory, PaymentMethod, OrderStatus
Key Numerical Fields	Quantity, UnitPrice, GrossSales, DiscountPct, NetSales, TotalAmount, CustomerRating
Temporal / Identifiers	OrderID, CustomerID, OrderDate, OrderDate.1, SalesRep

3. Project Repository Structure

```
Exploratory Data Analysis (EDA)/
├── salesdataset.csv           # Primary raw dataset
├── task5_eda.ipynb           # Execution notebook with all visualizations
├── task5_eda.pdf             # Exported report of analytical findings
└── README.pdf               # Project documentation & summary
```

4. Analysis Workflow & Visualization Roadmap

- **Structural Exploration:** Executed `df.info()`, `df.describe()`, and `value_counts()` to inspect schema, data types, null counts, and distribution spreads.
- **Univariate Distribution & Outlier Detection:** Created histograms and boxplots for `TotalAmount` and `CustomerRating` to assess skewness and extreme transaction values.
- **Bivariate & Categorical Comparison:** Evaluated `NetSales` grouped across 8 product categories using multi-colored boxplots.
- **Correlation Analysis:** Generated a full correlation matrix heatmap evaluating all floating-point and integer metrics.

- **Multivariate Interaction:** Explored pricing versus gross yield across categories via scatterplots and pairwise relationships via `sns.pairplot`.

5. Key Analytical Findings

Right-Skewed Revenue Distribution

Order amounts exhibit pronounced positive skewness. While the median order amount is **\$2,244.64**, the mean reaches **\$2,961.19**, pulled upward by bulk enterprise orders reaching up to **\$12,953.42**.

Balanced Sales Channels & Geographic Spread

Order volume is evenly distributed across all six sales channels (ranging between 2,477 on Amazon to 2,548 on eBay) and across continental regions (Africa, Asia, Europe, North America, Oceania, South America), proving low channel dependency risk.

Category Revenue Leadership

Books generated the highest net sales volume (**\$5.28M** across 1,896 orders), closely followed by **Office Supplies** (**\$5.15M** across 1,953 orders).

Discount Independence & Satisfaction Decoupling

`CustomerRating` demonstrates virtually zero correlation with `DiscountPct` ($r \approx 0.003$) and `TotalAmount` ($r \approx -0.020$). Discounts do not automatically yield higher review ratings.

6. Data Quality Issues Identified

- **Missing Values:** The `SalesRep` column has **1,696 missing entries** (~11.3% of 15,030 rows), typical for self-service digital web/app purchases.
- **Duplicate Entries:** Discovered **30 exact duplicate rows** in the raw dataset that require deduplication before training predictive algorithms.
- **Outliers:** Upper-bound outliers exist above the $1.5 \times IQR$ threshold on `TotalAmount` and `GrossSales`.

7. Recommendations & Next Steps

- **Data Cleaning:** Drop the 30 identical duplicate rows and impute missing `SalesRep` values with 'Unassigned / Digital'.
- **Feature Engineering:** Convert `OrderDate` to `datetime`; extract seasonal features (Quarter, Month, Day of Week); create binary flags for discounted orders.
- **Transformations:** Apply log-transformations or `RobustScaler` on monetary features before linear regression modeling.
- **Predictive Modeling:** Train an XGBoost / Random Forest Regressor to forecast order values, and implement Logistic Regression / LightGBM to predict cancellation risks.